SSE-KMS envelope encryption

Local 08 · 45–60 minute conceptual add-on after AWS Lab 5.1

PRIMARY LESSON

AWS Academy

Cloud Security Foundations Module 5 · Protecting data

COURSE ADD-ON

Local 08

Trace upload, read and KMS key-state effects.

Use the assigned AWS module and lab first; work on only your assigned Local Part.

Trace the key, not just the object

BASELINE

AWS Lab 5.1

Observe S3 encryption with KMS and relevant KMS events.

TODAY

Envelope flow

Separate the KMS key, plaintext data key and encrypted data key.

QUESTION

Later read

What must still be available to decrypt an old object?

Scenario: per-object SSE-KMS without S3 Bucket Keys.

Two copies of a data key

01

S3

Requests GenerateDataKey

→

02

KMS

Returns plaintext + encrypted copies

→

03

S3

Encrypts object with plaintext copy

→

04

Store

Ciphertext + encrypted data key

Use now

The plaintext data key encrypts this object.

Keep later

S3 stores the encrypted copy and discards the plaintext copy.

The KMS key protects the data key; it does not directly encrypt the whole object.

Why an old object needs KMS again

01

Stored object

Ciphertext + encrypted data key

→

02

S3

Sends encrypted data key to KMS

→

03

KMS

Decrypts it if key state and permissions allow

→

04

S3

Uses plaintext data key for this read

No plaintext cache

In this lesson scenario, S3 does not keep a plaintext data key between requests.

Access condition

Stored ciphertext alone does not grant a usable read.

Check the request path and key state when a read fails.

Reversible and irreversible outcomes

DISABLED

Temporarily unavailable

Existing ciphertext remains. Reads needing KMS fail until the key is re-enabled.

PENDING DELETION

Cancel before deadline

Cancellation stops deletion but leaves the key disabled; enable it again for use.

DELETED

Cannot recover key

Completed deletion is irreversible; ciphertext that depends on that key is unusable.

A key state change affects access; it does not erase the stored encrypted object.

One key for control, data keys for data

KMS KEY

Protects keys

A managed control point for authorization, audit and key state.

DATA KEY

Encrypts content

A fresh per-operation data key handles object data efficiently.

STORED COPY

Supports future reads

Only the encrypted data key is retained with the object.

Do not confuse one KMS key with one reusable plaintext data key for every object.

Audit a claim by tracing the flow

1 · NAME

Which key?

Identify whether the claim concerns the KMS key or a data key.

2 · TRACE

Upload or read?

Walk through the actual request and note what is stored or discarded.

3 · CORRECT

One precise sentence

Explain the key state and replace the false claim in your own words.

Section A: Part 1 · Section B: Part 2 · No flag is required for this conceptual task.